

SOC/L1 - 7.1. Network Security Essentials

Task 1 - Introduction

The networks are the backbone of every modern organization. Servers, workstations, applications, and security devices don't exist in isolation; they are interconnected, forming a single ecosystem. The network perimeter is where this internal ecosystem is segregated from the external Internet, and it is often the first target for attackers.

Task 2 - Lab Connection

All files required to complete the room are placed in the `Perimeter_logs` folder on the Desktop. The log files to be used in the task 7 for the investigation are pre-ingested into Splunk Instance.

Task 3 - A Network

User Workstations (Endpoints): Endpoint logs can reveal malicious processes, but network logs may first show C2 (Command & Control) connections.

File & Database Servers: File servers provide centralized access to shared documents, while database servers manage structured data like customer records, HR information, or financial data. Importance: Attackers usually target these servers because compromising them means access to valuable or sensitive data. Ransomware operators, in particular, target file servers to maximize their impact. Data exfiltration campaigns often involve stealthily moving data from these servers out of the network.

Application Servers: Web, Email, VPN, etc.

- Web Servers: Host company websites and web applications.
- Email Servers: Handle corporate communications.
- VPN Gateways: Allow secure remote access to internal resources.

Importance: Attackers scan them constantly for software vulnerabilities or weak configurations. Exploiting one often provides a foothold into the internal network. From a security perspective, we need to monitor application logs, firewall alerts, and IDS signatures to identify:

- Exploit attempts (e.g., SQL injection on a web app).
- Brute-force login attempts on email or VPN services.
- Suspicious external IPs interacting with sensitive applications.

Active Directory (AD) / Authentication Servers: Active Directory is the identity backbone of most enterprise networks. It manages users, groups, computers, and their access rights. Employees use their AD credentials to log into their computers, access email, file servers, and internal applications.

- AD is the main component that controls all user accounts and systems within the network.
- Attackers commonly target for privilege escalation, AD, persistence and lateral movement.
- A single compromised domain admin account can bring down the entire enterprise.

From a security perspective, we need to monitor authentication logs for suspicious behavior such as:

- Multiple failed login attempts (password spraying).
- Unusual logins from external IPs or at odd hours.
- Accounts accessing systems they normally shouldn't.

Routers & Switches:

Routers connect different networks, most importantly linking the enterprise LAN to the Internet. Switches connect devices within the same network, ensuring employees' PCs, printers, and servers communicate seamlessly. Importance: While routers and switches are not directly exposed in most enterprise setups, if compromised, they provide attackers with the ability to:

- Intercept and manipulate traffic (Man-in-the-Middle attacks).
- Create backdoors by rerouting traffic.

- Open hidden channels to the Internet.

Firewalls / Perimeter Devices: A firewall is the primary security gateway that controls traffic between the trusted internal network and the untrusted Internet. It inspects incoming and outgoing packets and decides whether to allow or block them based on defined security rules. Modern firewalls also perform deep inspection of applications, intrusion prevention, and even malware detection. Importance:

- Protects the enterprise from direct exposure to the Internet.
- Prevents unauthorized access to internal services (e.g., database or RDP).
- Logs every connection attempt, successful or blocked. These logs are often the earliest indicators of attacks such as port scans, brute-force attempts, or exploitation attempts.

Task 4 - Network Visibility

Network visibility allows analysts to monitor and understand network activity, which is essential for detecting threats, investigating incidents, and maintaining a strong security posture. The network visibility comes from two primary sources of logs. Effective visibility allows security analysts to:

- **Detect Anomalies:** Spot unusual patterns that could indicate an attack.
- **Investigate Incidents:** Piece together the events of an attack to understand what happened.
- **Hunt for Threats:** Proactively search for hidden adversaries in the network.
- **Ensure Compliance:** Meet regulatory requirements by logging and monitoring network activity.

Host-Centric Logs: Host-centric logs record internal activity on individual endpoint devices—like servers and workstations—providing a detailed look at how events impact a specific machine. Key Host-Centric Log Sources:

- **Operating System Logs:** Windows Event Logs, Linux `syslog`, macOS logs. These record events like user logons, process creation, service startups, and failed login attempts.

- **Application Logs:** Logs from software running on the host, such as web servers (, Apache Nginx), databases (MySQL, MSSQL), and other applications.
- **Security Tool Logs:** Logs from antivirus software, Endpoint Detection and Response (EDR) agents, and host-based intrusion detection systems (HIDS).

These logs are invaluable for answering questions like:

- **Detailed Forensic Analysis:** Understanding the exact actions an attacker took on a compromised machine, such as which files were accessed, modified, or deleted.
- **Process and Execution Tracking:** Identifying the creation of malicious processes, the execution of unauthorized scripts (like PowerShell), and changes to system services.
- **User Activity Monitoring:** Tracking who logged in, when they logged in, and what privileges they used. This is essential for detecting both external attacks and insider threats.
- **Malware Impact Assessment:** Confirming if a malicious file was executed and what changes it made to the system registry, file system, or running services.

Network-Centric Logs: While host logs tell us what's happening on a device, network-centric logs tell us what's happening between devices. These are generated by network appliances that sit on the network and monitor the traffic flowing through them.

- **What they show:** Source and destination IPs, ports, protocols, and the action taken (e.g, allowed or blocked).
- **Why they are important:** They give you the crucial "when," "where," and "how." They can reveal an attacker's initial reconnaissance attempts, lateral movement between systems, or attempts to exfiltrate data.

Key Network-Centric Log Sources

- **Firewalls:** The firewall logs provide a record of every connection allowed or denied based on pre-defined security rules. Firewall logs are the first place to look for unauthorized connection attempts from the internet.
- **Intrusion Detection/Prevention Systems (IDS/IPS):** They monitor network traffic for patterns that match known malicious attacks (signatures) or for

unusual behavior (anomalies). Their logs are critical for detecting active attacks in real time.

- **Routers and Switches:** While not logging in the traditional sense, devices like routers can generate flow data. This data summarizes traffic conversations, telling who talked to whom, for how long, and how much data was sent. It's excellent for getting a high-level overview of network activity.
- **Web Proxies:** These logs are a goldmine for organizations that route their web traffic through proxies. They record every website user's visit, providing visibility into web-based threats, policy violations, or data exfiltration attempts.
- **VPN:** These devices manage remote access for employees. Their logs track who is connecting to the corporate network, from where, and at what time, which is essential for monitoring the security of remote connections.

Importance of Network-Centric Logs

They are essential for:

- **Early Threat Detection:** Identifying threats at the network edge before they can compromise an endpoint. This includes blocking port scans, brute-force attempts, and connections from known malicious IPs.
- **Identifying Command & Control (C2):** Spotting communication patterns between a compromised internal host and an external attacker-controlled server.
- **Tracking Lateral Movement:** Observing an attacker moving from one compromised machine to another within the internal network.
- **Detecting Data Exfiltration:** Alerting on unusually large or suspicious outbound data transfers that could signal a data breach.
- **Providing Broad Context:** Understanding the scope of an attack by seeing which other devices a compromised host tried to communicate with.

Task 5 - Network Perimeter

The network perimeter acts as the boundary between a trusted internal network and the untrusted internet, serving as the primary entry point and first line of defense where all incoming and outgoing traffic is monitored and controlled.

The Perimeter: The perimeter is defined by hardware devices at the edge of the network. However, it also includes virtual gateways, cloud connections, and remote access points in modern environments. Common components of a network perimeter include:

- Firewalls: Gatekeepers that filter traffic between internal and external networks.
- Routers/Gateways: Devices that route traffic and enforce access rules.
- Demilitarized Zone (DMZ): A buffer network segment where public-facing servers (web, mail, VPN) are placed.
- Remote Access Gateways / VPNs: Secure entry points for employees working outside the office.

Importance of Network Perimeter

The network perimeter serves as the primary line of defense, using combined security controls to filter traffic, protect internal assets, and expose early signs of external attack. If the perimeter is weak or misconfigured, attackers can:

- Exploit exposed services (e.g., RDP , MySQL, SMB) to gain access.
- Perform scanning and reconnaissance to map the network.
- Launch brute-force attacks against login services.
- Use data exfiltration channels to send stolen data out.

The perimeter is the first line of defense against external threats. Attackers scan perimeter IPs looking for open ports and vulnerabilities. Misconfigured or weakly protected perimeters often lead to:

- Unauthorized access (e.g.,exposed RDP /SSH)
- Data breaches
- Malware infiltration

Role of a Security Analyst: As a security analyst, monitoring the perimeter means:

- Reviewing firewall logs for blocked/allowed connections.
- Identifying scanning attempts or brute-force login attempts.

- Flagging unusual outbound traffic that may indicate malware beaconing or exfiltration.
- Understanding what should (and should not) be exposed at the perimeter.

Task 6 - Network Perimeters: Monitoring and Protecting

Monitoring the perimeter means using firewalls, intrusion detection/prevention systems (IDS/IPS), and access control to examine and limit exposure and enforce security rules. This allows the security analysts to:

- Spot early-stage attacks like port scans or brute-force attempts.
- Detect misconfigurations that leave sensitive services exposed.
- Identify outbound traffic anomalies that may indicate malware or data exfiltration.

Scenario 1: Probing for Ports (Port Scanning): An attacker is testing your network to see which ports are open or closed, while the firewall is doing its job and blocking them.

Scenario 2: Attacking the Web Server (SQL Injection): The company website is being attacked. An intrusion detection system (IDS) provides more details than a firewall, identifying the type of attack. Investigate WAF (Web Application Firewall) Logs

Scenario 3: Guessing the Password (VPN Brute-Force): An attacker is trying to guess a user's password to gain remote access to the network. Investigate VPN Gateway Log

Answer the questions below

Examine the firewall logs. Which IP address is performing the port scan?

203.0.113.10 (Go to task 6, then run the query: `grep -i block head firewall_logs.txt`)

```
firewall_logs.txt:2025-09-15 08:30:05 BLOCK TCP 203.0.113.10:50001 -> 10.0.0.20:21
firewall_logs.txt:2025-09-15 08:30:06 BLOCK TCP 203.0.113.10:50002 -> 10.0.0.20:22
firewall_logs.txt:2025-09-15 08:30:08 BLOCK TCP 203.0.113.10:50003 -> 10.0.0.20:23
firewall_logs.txt:2025-09-15 08:30:09 BLOCK TCP 203.0.113.10:50004 -> 10.0.0.20:25
firewall_logs.txt:2025-09-15 08:30:11 BLOCK TCP 203.0.113.10:50005 -> 10.0.0.20:53
firewall_logs.txt:2025-09-15 08:30:13 BLOCK TCP 203.0.113.10:50006 -> 10.0.0.20:80
```

In the WAF Logs, which single source IP is responsible for all the blocked web attacks?

198.51.100.12 (run the query: `grep -i block head waf_logs.txt`)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/task6$ grep -i block head waf_logs.txt
```

```
waf_logs.txt:timestamp=2025-09-22T09:14:10Z src_ip=198.51.100.12 action=BLOCK request="GET /products.php?id=12%20UNION%20SELECT%20user,pass%20FROM%20users" rule_id=942100 attack_type="SQL Injection"
```

In the VPN logs, how many brute-force attempts failed?

90 (run the query: `grep FAILED vpn_logs.txt | wc -l`)

Which suspicious IP address was found attempting the brute-force attack against the VPN gateway? 45.137.22.13

Task 7 - Perimeter Logs: Investigating the Breach

Incident Scenario: Initech Corp, a mid-sized financial services company, has recently deployed a new firewall and intrusion detection system (IDS) to monitor its network perimeter. Over the past month, security analysts have noticed abnormal traffic patterns, but the SOC team has been overwhelmed and missed deeper analysis.

- **Firewall Logs:** `firewall.log`
- **WAF Logs:** `ids_alerts.log`
- **VPN Logs:** `vpn_auth.log`

Investigating the Logs: There are two ways to investigate the logs: manually using command-line tools or using Splunk.

Method 1: Manual Log Analysis:

`head firewall.log` (displays the first 10 lines of the firewall.log file.)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ head firewall.log
2025-08-25 00:47:46 ALLOW TCP [REDACTED]:60317 -> 10.0.0.50:443
2025-08-25 01:29:33 ALLOW TCP 203.0.113.100:62718 -> [REDACTED]:443
2025-08-25 01:42:12 ALLOW TCP 203.0.113.100:55875 -> [REDACTED]:80
```

`head ids_alerts.log` (displays the first 10 lines of the ids_alerts.log file.)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ head ids_alerts.log
2025-08-25 00:12:53 [**] [1:2003272:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activity] [Priority: 3] {TCP} 198.51.100.92:20127 -> [REDACTED]:22
2025-08-25 01:50:30 [**] [1:2003377:1] ET POLICY Suspicious HTTP [**] [Classification: Suspicious Activity] [Priority: 1] {TCP} 203.0.113.100:56603 -> [REDACTED]:25
```

`head vpn_auth.log` (displays the first 10 lines of the vpn_auth.log file.)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ head vpn_auth.log
2025-08-25 08:25:10 [REDACTED] alice SUCCESS assigned_ip=10.8.0.143
2025-08-25 08:27:38 203.0.113.100 svc_[REDACTED] SUCCESS assigned_ip=10.8.0.131
2025-08-25 14:57:10 203.0.113.10 svc_[REDACTED] SUCCESS assigned_ip=10.8.0.116
```

Reconnaissance attempt:

`cat firewall.log | grep "BLOCK" | head` analyzing the blocked requests in the firewall logs

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep "BLOCK" | head
2025-08-26 12:12:47 BLOCK TCP 203.0.113.10:64292 -> 10.0.0.50:21
2025-08-27 03:18:28 BLOCK TCP [REDACTED]:61701 -> [REDACTED]:23
2025-08-27 11:56:20 BLOCK TCP 203.0.113.10:64952 -> 10.0.0.50:22
```

`cat firewall.log | grep "BLOCK" | cut -d' ' -f5 | cut -d: -f1 | sort -nr | uniq -c` (which IP is responsible for the maximum BLOCK requests.)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep "BLOCK" | cut -d' ' -f5 | cut -d: -f1 | sort -nr | uniq -c
279 [REDACTED]
46 203.0.113.10
26 [REDACTED]
```

`cat firewall.log | grep [REDACTED] | grep "ALLOW"` (if our firewall has allowed any requests to the suspicious IP)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep [REDACTED] | grep "ALLOW"
2025-08-26 00:17:58 ALLOW TCP [REDACTED]:61009 -> [REDACTED]:4444
2025-08-26 22:04:34 ALLOW TCP [REDACTED]:55996 -> 10.0.0.50:445
2025-08-27 21:04:23 ALLOW TCP [REDACTED]:53944 -> 10.0.0.50:22
```

VPN Brute-force / Credential Access

`cat vpn_auth.log | grep FAIL | cut -d' ' -f3 | sort -nr | uniq -c` (the number of failed attempts)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat vpn_auth.log | grep FAIL | cut -d' ' -f3 | sort -nr | uniq -c
118 [REDACTED]
1 203.0.113.100
1 198.51.100.92
1 198.51.100.45
```

`cat vpn_auth.log | grep [REDACTED]` (multiple login attempt was made against a certain user)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat vpn_auth.log | grep [REDACTED]
2025-09-03 02:19:00 [REDACTED] svc_[REDACTED] FAIL
2025-09-03 02:19:10 [REDACTED] svc_[REDACTED] FAIL
2025-09-03 02:19:20 [REDACTED] svc_[REDACTED] FAIL
```

Lateral Movement

`cat firewall.log | grep [REDACTED] | grep "ALLOW" | head` (filters the firewall log for a specific IP or event, isolates permitted connections (ALLOW), and prints the first 10 matching entries to analyze successful traffic patterns.)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep [REDACTED] | grep "ALLOW" | head
2025-09-05 06:00:00 ALLOW TCP [REDACTED]:2000 -> [REDACTED]:2
2
2025-09-05 06:10:00 ALLOW TCP [REDACTED]:2001 -> [REDACTED]:4
45
2025-09-05 06:20:00 ALLOW TCP [REDACTED]:2002 -> [REDACTED]:2
2
```

`cat ids_alerts.log | grep [REDACTED] | head` (filter through the compromised IP and see if we can find any intrusion rules triggered)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat ids_alerts.log | grep [REDACTED] | head
```

```
2025-09-05 06:00:00 [**] [1:2000200:1] ET SCAN Possible SSH S
can [**] [Classification: Attempted Unauthorized Access] [Pri
ority: 1] {TCP} [REDACTED]:2000 -> [REDACTED]:22
2025-09-05 06:10:00 [**] [1:2000201:1] ET EXPLOIT Possible MS
-SMB Lateral Movement [**] [Classification: Attempted Unautho
rized Access] [Priority: 1] {TCP} [REDACTED]:2001 -> [REDACTE
D]:445
```

```
cat ids_alerts.log | grep -n [REDACTED] | grep 'SMB' | cut -d' ' -f6,7,8,9,10,19,21 | head
```

(multiple services across the network, with an IDS alert highlighting a potential SMB exploit)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat ids_
alerts.log | grep -n [REDACTED] | grep 'SMB' | cut -d' ' -f6,
7,8,9,10,19,21 | head
EXPLOIT Possible MS-SMB Lateral Movement [REDACTED]:2001 [RED
ACTED]:445
EXPLOIT Possible MS-SMB Lateral Movement [REDACTED]:2006 [RED
ACTED]:445
EXPLOIT Possible MS-SMB Lateral Movement [REDACTED]:2010 [RED
ACTED]:445
```

Server Message Block (SMB) is a network protocol used mainly in Windows environments for sharing files, printers, and system resources between devices over TCP port 445.

C2 Beaconsing

C2 Beaconsing is the continuous, automated outbound communication sent from malware on a compromised host to an attacker's server to check in and receive instructions.

```
cat ids_alerts.log | grep C2 | head
```

 (the initial 10 Command and Control (C2) detection alerts from the IDS log to quickly identify infected hosts and external attacker infrastructure.)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat ids_
alerts.log | grep C2 | head
```

```
2025-09-11 01:00:00 [**] [1:2001000:1] ET TROJAN Possible C2  
Beaconing [**] [Classification: A network Trojan was detecte  
d] [Priority: 1] {TCP} [REDACTED]:30000 -> [REDACTED]:4444
```

```
cat ids_alerts.log | grep -n [REDACTED] | cut -d' ' -f6,7,8,9,10,19,22,23 | head -n 15
```

(filter against the compromised IP responsible for C2 beaconing)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat ids_  
alerts.log | grep -n [REDACTED] | cut -d' ' -f6,7,8,9,10,1  
9,22,23 | head -n 15  
POLICY Suspicious HTTP [**] [Classification:  
WEB_SERVER Possible SQL Injection [**] [REDACTED]:3389
```

```
cat ids_alerts.log | grep -n [REDACTED] | cut -d' ' -f6,7,8,9,10,19,22,23 | uniq -c | sort -  
nr | head
```

(This command aggregates and counts specific fields from matched IDS alerts, sorting them to reveal the most frequent threat activity for a target host.)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat ids_  
alerts.log | grep -n [REDACTED] | cut -d' ' -f6,7,8,9,10,19,2  
2,23 | uniq -c | sort -nr | head  
32 TROJAN Possible C2 Beaconing [**] {TCP} [REDACTED]:4444  
5 POLICY Suspicious HTTP [**] [Classification:
```

```
cat ids_alerts.log | grep -n [REDACTED] | cut -d' ' -f6,7,8,9,10,19,22,23 | uniq -c | sort  
-nr | head
```

(This is the exact command from your previous turn—it filters, slices, groups, and ranks the most frequent IDS alerts for that target host.)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat ids_  
alerts.log | grep -n [REDACTED] | cut -d' ' -f6,7,8,9,10,19,  
22,23 | sort -nr | uniq -c | sort -nr  
80 TROJAN Possible C2 Beaconing [**] {TCP} [REDACTED]:4444  
32 INFO Possible HTTP POST Large {TCP} [REDACTED]:80  
28 INFO Possible HTTP POST Large {TCP} [REDACTED]:8080  
23 POLICY Suspicious HTTP [**] [Classification:
```

Data Exfiltration Attempt

```
cat firewall.log | grep [REDACTED] | cut -d' ' -f5,6,7 | uniq | sort
```

 (filter on the compromised hosts, and examine the traffic originating from those to an external destination IP)

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep [REDACTED] | cut -d' ' -f5,6,7 | uniq | sort
[REDACTED]:40000 -> [REDACTED]:8080
[REDACTED]:40001 -> [REDACTED]:8080
[REDACTED]:40002 -> [REDACTED]:8080
[REDACTED]:40003 -> [REDACTED]:80
[REDACTED]:40004 -> [REDACTED]:80
```

Method 2: Analyzing Logs via Splunk.

To proceed, open the link `localhost:8000` in the browser, click on the **Search & Reporting** tab on the left bar, and start analyzing the logs. Logs are pre-ingested into the `index="network_logs"`, as shown below:

New Search

index="network_logs"

✓ 2,485 events (before 9/24/25 11:17:23.000 AM) No Event Sampling ▾

Events (2,485) Patterns Statistics Visualization

Timeline format ▾ - Zoom Out + Zoom to Selection × Deselect

< Hide Fields ≡ All Fields

SELECTED FIELDS

- a host 1
- a source 3
- a sourcetype 3

INTERESTING FIELDS

- a action 2
- a alert 10
- a classification 6
- # date_hour 24
- # date_mday 31
- # date_minute 60
- a date_month 2
- # date_second 60

sourcetype

3 Values, 100% of events

Selected Yes No

Reports

- Top values
- Top values by time
- Rare values

Events with this field

Values	Count	%
firewall_logs	1,284	51.67%
ids_logs	983	39.557%
vpn_logs	218	8.773%

"priority": 2,
Show all 13 lines

host = perimeter_logs source = ids_alerts.json sourcetype = ids_logs

FIREWALL Investigation:

1. Select the action: Allow or Block

```
index="network_logs" sourcetype=firewall_logs action=BLOCK
| table _time src_ip src_port dst_ip dst_port
| sort -_time
```

This SPL query retrieves blocked firewall connections, formats the logs into a clean table showing timestamp, source, and destination details, and displays them from newest to oldest.

New Search Save As ▾ Create Table View Close

```
1 index="network_logs" sourcetype=firewall_logs action=BLOCK
2 | table _time src_ip src_port dst_ip dst_port
3 | sort -_time
```

Time range: All time ▾ 🔍

✓ 351 events (before 7/29/26 8:38:28.000 AM) No Event Sampling ▾ Job ▾ ⏏ ⏏ ⏏ ⏏ ⏏ ⏏ ⏏ Smart Mode ▾

Events Patterns **Statistics (351)** Visualization

Show: 20 Per Page ▾ Format ▾ Preview: On < Prev 1 2 3 4 5 6 7 8 ... Next >

_time ▾	src_ip ▾	src_port ▾	dst_ip ▾	dst_port ▾
2025-09-22 19:58:18	203.0.113.45	52946	10.0.0.20	21
2025-09-22 11:42:08	203.0.113.10	56373	10.0.0.20	445
2025-09-22 11:16:52	203.0.113.10	63010	10.0.0.60	23
2025-09-22 08:23:47	203.0.113.45	50984	10.0.0.50	23
2025-09-21 19:15:37	203.0.113.10	50289	10.0.0.60	4444
2025-09-21 12:56:37	203.0.113.10	51359	10.0.0.20	23
2025-09-21 10:05:28	203.0.113.10	59559	10.0.0.20	22
2025-09-21 08:04:09	203.0.113.10	50395	10.0.0.50	445
2025-09-21 06:38:36	203.0.113.10	50691	10.0.0.51	4444
2025-09-21 03:24:53	203.0.113.10	55526	10.0.0.51	22

2. This SPL query aggregates blocked firewall events by destination IP and ranks them in descending order to identify the most heavily targeted external or internal destinations.

```
index="network_logs" sourcetype=firewall_logs action=BLOCK
| stats count by dst_ip
| sort -count
```

New Search Save As ▾ Create Table View Close

```

1 index="network_logs" sourcetype=firewall_logs action=BLOCK
2 | stats count by dst_ip
3 | sort -count

```

Time range: All time 🔍

✓ 351 events (before 7/29/26 8:40:59.000 AM) No Event Sampling ▾ Job ▾ || ▢ → 🖨️ ⬇️ 💡 Smart Mode ▾

Events Patterns **Statistics (4)** Visualization

Show: 20 Per Page ▾ ✍️ Format ▾ 🔵 Preview: On

dst_ip ↕	count ↕
10.0.0.20	145
10.0.0.50	145
10.0.0.60	34
10.0.0.51	27

3. This SPL query aggregates blocked firewall events by source IP and ranks them in descending order to identify the top hosts generating rejected traffic.

```

index="network_logs" sourcetype=firewall_logs action=BLOCK
| stats count, by src_ip
| sort -count

```

New Search Save As ▾ Create Table View Close

```

1 index="network_logs" sourcetype=firewall_logs action=BLOCK
2 | stats count, by src_ip
3 | sort -count

```

Time range: All time 🔍

✓ 351 events (before 7/29/26 8:43:57.000 AM) No Event Sampling ▾ Job ▾ || ▢ → 🖨️ ⬇️ 💡 Smart Mode ▾

Events Patterns **Statistics (3)** Visualization

Show: 20 Per Page ▾ ✍️ Format ▾ 🔵 Preview: On

src_ip ↕	count ↕
203.0.113.45	279
203.0.113.10	46
10.8.0.23	26

VPN Logs Investigation:

1. This SPL query counts failed VPN logins per source IP and returns the single address responsible for the highest number of failed authentication attempts.

```
index="network_logs" sourcetype="vpn_logs"
result : fail or success
src_ip: most failed ip
```

2. This SPL query lists the timestamps, targeted usernames, and assigned IP fields for all failed VPN login attempts originating specifically from source IP

203.0.113.45 .

```
index="network_logs" sourcetype="vpn_logs" result=FAIL src_ip=
"203.0.113.45"
| table _time username assigned_ip
```

New Search Save As ▾

```
1 index="network_logs" sourcetype="vpn_logs" result=FAIL src_ip="203.0.113.45"
2 | table _time username assigned_ip
```

✓ 118 events (before 7/29/26 8:53:41.000 AM) No Event Sampling ▾ Job ▾ || ▣ ➔

Events Patterns **Statistics (118)** Visualization

Show: 20 Per Page ▾ Format ▾ Preview: On < Prev 1

<u>_time</u> ↕	<u>username</u> ↕	<u>assigned_</u>
2025-09-03 02:09:30	svc_backup	null
2025-09-03 02:09:20	svc_backup	null
2025-09-03 02:09:10	svc_backup	null
2025-09-03 02:09:00	svc_backup	null
2025-09-03 02:08:50	svc_backup	null
2025-09-03 02:08:40	svc_backup	null
2025-09-03 02:08:30	svc_backup	null

3. This SPL query extracts VPN log details into a clean table containing the timestamp, source IP, username, and assigned internal IP, displaying them in chronological order from oldest to newest.

```
index="network_logs" sourcetype=vpn_logs
| table _time src_ip username assigned_ip
| sort _time
```

New Search

Save As Create Table View Close

```
1 index="network_logs" sourcetype=vpn_logs
2 | table _time src_ip username assigned_ip
3 | sort _time
```

Time range: All time



✓ 218 events (before 7/29/26 10:08:54.000 AM) No Event Sampling

Job II Smart Mode

Events Patterns **Statistics (218)** Visualization

Show: 20 Per Page Format Preview: On < Prev 1 ... 4 5 6 7 8 9 10 11 Next >

_time	src_ip	username	assigned_ip
2025-09-20 00:01:21	198.51.100.92	jsmith	null
2025-09-22 09:37:02	203.0.113.100	jsmith	10.8.0.199
2025-09-09 15:22:15	198.51.100.45	svc_backup	10.8.0.198
2025-09-14 06:49:06	198.51.100.77	jsmith	10.8.0.197

IDS_Logs

chose alert: ET EXPLOIT Possible MS-SMB Lateral Movement

1. This SPL query filters IDS logs for "Possible MS-SMB Lateral Movement" alerts triggered by source IP 10.8.0.23, displaying the destination IP and port targets in reverse chronological order.

```
index="network_logs" sourcetype="ids_logs" src_ip="10.8.0.23"
alert="ET EXPLOIT Possible MS-SMB Lateral Movement"
| table _time dst_ip dst_port
| sort -_time
```

New Search

Save As Create Table View Close

1 index="network_logs" sourcetype="ids_logs" src_ip="10.8.0.23" alert="ET EXPLOIT Possible MS-SMB Lateral Movement"
2 | table _time dst_ip dst_port
3 | sort -_time

Time range: All time

Q

32 events (before 7/29/26 10:20:30.000 AM) No Event Sampling Job ||

Smart Mode

Events Patterns Statistics (32) Visualization

Show: 20 Per Page Format Preview: On < Prev 1 2 Next >

_time	dst_ip	dst_port
2025-09-05 20:50:00	10.0.0.51	445
2025-09-05 20:30:00	10.0.0.51	445
2025-09-05 19:40:00	10.0.0.20	445
2025-09-05 18:50:00	10.0.0.51	445
2025-09-05 18:30:00	10.0.0.60	445
2025-09-05 18:20:00	10.0.0.20	445
2025-09-05 18:00:00	10.0.0.51	445

chose alert: alert="ET TROJAN Possible C2 Beaconsing

2. This SPL query displays reverse-chronological connection details for "Possible C2 Beaconsing" alerts originating from source IP 10.0.0.60 to identify active malware check-ins and external command servers.

```
index="network_logs" sourcetype="ids_logs" alert="ET TROJAN P
ossible C2 Beaconsing" src_ip="10.0.0.60"
| table _time dst_ip dst_port src_ip src_port
| sort -_time
```

New Search Save As Create Table View Close

```

1 index="network_logs" sourcetype="ids_logs" alert="ET TROJAN Possible C2 Beaconsing" src_ip="10.0.0.60"
2 | table _time dst_ip dst_port src_ip src_port
3 | sort -_time

```

Time range: All time Q

✓ 80 events (before 7/29/26 10:25:49.000 AM) No Event Sampling Job || ≡ ↗ 📄 ⬇ 💡 Smart Mode

Events Patterns **Statistics (80)** Visualization

Show: 20 Per Page Format Preview: On < Prev 1 2 3 4 Next >

_time	dst_ip	dst_port	src_ip	src_port
2025-09-26 03:00	198.51.100.77	4444	10.0.0.60	30063
2025-09-26 03:00	198.51.100.77	4444	10.0.0.60	30062
2025-09-26 03:00	198.51.100.77	4444	10.0.0.60	30061
2025-09-26 01:00	198.51.100.77	4444	10.0.0.60	30064
2025-09-26 01:00	198.51.100.77	4444	10.0.0.60	30060
2025-09-25 19:00	198.51.100.77	4444	10.0.0.60	30059
2025-09-25 13:00	198.51.100.77	4444	10.0.0.60	30058
2025-09-25 07:00	198.51.100.77	4444	10.0.0.60	30065

Answer the questions below

Examine the firewall logs. What external IP performed the most reconnaissance?

203.0.113.45

run the query: `grep BLOCK firewall.log | grep "BLOCK" | cut -d' ' -f5 | cut -d: -f1 | sort -nr | uniq -c`

```

ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ grep BLOCK firewall.log
| grep "BLOCK" | cut -d' ' -f5 | cut -d: -f1 | sort -nr | uniq -c
  279 203.0.113.45
   46 203.0.113.10
   26 10.8.0.23

```

run the query: `grep FAIL vpn_auth.log | cut -d' ' -f3 | sort -nr | uniq -c`

`cut -d' ' -f5`: Extracts the 5th space-delimited field containing the IP and port (e.g., 10.0.0.60:443).

`cut -d: -f1`: Strips the colon and port number to isolate just the IP address (10.0.0.60).

`cut -d' ' -f3`: Splits the line by spaces (`-d' '`) and extracts the 3rd field (`-f3`).

`sort -nr`: Sorts items numerically in reverse order.

`uniq -c`: Groups adjacent duplicate lines and outputs occurrence counts.

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ grep FAIL vpn_auth.log | cut -d' ' -f3 | sort -nr | uniq -c
    118 203.0.113.45
      1 203.0.113.100
      1 198.51.100.92
      1 198.51.100.45
```

In the firewall log, Which internal host was targeted by scans?

10.0.0.20

```
grep "BLOCK" firewall.log | head
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$
grep "BLOCK" firewall.log | head
2025-08-26 12:12:47 BLOCK TCP 203.0.113.10:64292 -> 10.0.0.50:21
2025-08-27 03:18:28 BLOCK TCP 203.0.113.45:61701 -> 10.0.0.60:23
2025-08-27 11:56:20 BLOCK TCP 203.0.113.10:64952 -> 10.0.0.50:22
2025-08-27 22:52:00 BLOCK TCP 203.0.113.10:63686 -> 10.0.0.20:445
2025-08-28 10:00:00 BLOCK TCP 203.0.113.45:50000 -> 10.0.0.20:4444
2025-08-28 10:00:30 BLOCK TCP 203.0.113.45:50001 -> 10.0.0.20:21
2025-08-28 10:01:00 BLOCK TCP 203.0.113.45:50002 -> 10.0.0.20:21
2025-08-28 10:01:30 BLOCK TCP 203.0.113.45:50003 -> 10.0.0.20:4444
2025-08-28 10:02:00 BLOCK TCP 203.0.113.45:50004 -> 10.0.0.20:23
2025-08-28 10:02:30 BLOCK TCP 203.0.113.45:50005 -> 10.0.0.20:22
```

Which username was targeted in VPN logs?

svc_backup

```
grep -E "203.*FAIL" vpn_auth.log
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ grep -E "203.*FAIL" vpn_auth.log
2025-09-03 02:00:00 203.0.113.45 svc_backup FAIL
2025-09-03 02:00:10 203.0.113.45 svc_backup FAIL
2025-09-03 02:00:20 203.0.113.45 svc_backup FAIL
2025-09-03 02:00:30 203.0.113.45 svc_backup FAIL
2025-09-03 02:00:40 203.0.113.45 svc_backup FAIL
```

What internal IP was assigned after successful VPN login?

10.8.0.23

```
grep -E "svc_backup.*SUCCESS" vpn_auth.log
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ grep -E "svc_backup.*SUCCESS" vpn_auth.log
25-08-25 08:27:38 203.0.113.100 svc_backup SUCCESS assigned_ip=10.8.0.131
25-08-25 14:57:10 203.0.113.10 svc_backup SUCCESS assigned_ip=10.8.0.116
25-08-26 03:36:17 198.51.100.92 svc_backup SUCCESS assigned_ip=10.8.0.62
25-08-26 10:02:45 198.51.100.92 svc_backup SUCCESS assigned_ip=10.8.0.81
25-08-28 03:20:33 203.0.113.45 svc_backup SUCCESS assigned_ip=10.8.0.193
25-08-28 04:54:52 203.0.113.45 svc_backup SUCCESS assigned_ip=10.8.0.104
25-08-31 12:25:56 192.0.2.115 svc_backup SUCCESS assigned_ip=10.8.0.20
```

```
grep -E "ALLOW.*10.8.0.23" firewall.log
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ grep -E "ALLOW.*10.8.0.23" firewall.log
2025-09-05 06:00:00 ALLOW TCP 10.8.0.23:2000 -> 10.0.0.51:22
2025-09-05 06:10:00 ALLOW TCP 10.8.0.23:2001 -> 10.0.0.51:445
2025-09-05 06:20:00 ALLOW TCP 10.8.0.23:2002 -> 10.0.0.60:22
2025-09-05 06:40:00 ALLOW TCP 10.8.0.23:2004 -> 10.0.0.51:3389
2025-09-05 06:50:00 ALLOW TCP 10.8.0.23:2005 -> 10.0.0.20:22
2025-09-05 07:00:00 ALLOW TCP 10.8.0.23:2006 -> 10.0.0.20:445
2025-09-05 07:10:00 ALLOW TCP 10.8.0.23:2007 -> 10.0.0.20:22
```

Which port was used for lateral SMB attempts?

445

```
grep -E "SMB.*10.8.0.23" ids_alerts.log | cut -d' ' -f6,7,8,9,10,19,21 | head
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ grep -E "SMB.*10.8.0.23" ids_alerts.log | cut -d' ' -f6,7,8,9,10,19,21 | head
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2001 10.0.0.51:445
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2006 10.0.0.20:445
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2010 10.0.0.60:445
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2016 10.0.0.60:445
EXPLOIT Possible MS-SMB Lateral Movement 10.8.0.23:2018 10.0.0.20:445
```

In the IDS logs, which host beaconsed to the C2?

10.0.0.60

During the investigation, which IP was observed to be associated with C2?

198.51.100.77

```
grep C2 ids_alerts.log
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ grep C2 ids_alerts.log
2025-09-11 01:00:00 [**] [1:2001000:1] ET TROJAN Possible C2 Beaconsing [**] [Classified]
rk Trojan was detected] [Priority: 1] {TCP} 10.0.0.60:30000 -> 198.51.100.77:4444
2025-09-11 07:00:00 [**] [1:2001001:1] ET TROJAN Possible C2 Beaconsing [**] [Classified]
rk Trojan was detected] [Priority: 1] {TCP} 10.0.0.60:30001 -> 198.51.100.77:4444
2025-09-11 13:00:00 [**] [1:2001002:1] ET TROJAN Possible C2 Beaconsing [**] [Classified]
rk Trojan was detected] [Priority: 1] {TCP} 10.0.0.60:30002 -> 198.51.100.77:4444
2025-09-11 19:00:00 [**] [1:2001003:1] ET TROJAN Possible C2 Beaconsing [**] [Classified]
rk Trojan was detected] [Priority: 1] {TCP} 10.0.0.60:30003 -> 198.51.100.77:4444
2025-09-12 01:00:00 [**] [1:2001004:1] ET TROJAN Possible C2 Beaconsing [**] [Classified]
rk Trojan was detected] [Priority: 1] {TCP} 10.0.0.60:30004 -> 198.51.100.77:4444
```

Which host showed the exfiltration attempts?

10.0.0.51

```
grep "198.51.100.77" firewall.log
```

```
2025-09-13 21:43:36 ALLOW TCP 198.51.100.77:60129 -> 10.0.0.51:80
2025-09-13 23:44:10 ALLOW TCP 198.51.100.77:54871 -> 10.0.0.50:8080
2025-09-14 00:24:33 ALLOW TCP 198.51.100.77:59133 -> 10.0.0.20:80
2025-09-14 01:00:00 ALLOW TCP 10.0.0.60:30012 -> 198.51.100.77:4444
2025-09-14 03:26:34 ALLOW TCP 198.51.100.77:58420 -> 10.0.0.51:80
2025-09-14 07:00:00 ALLOW TCP 10.0.0.60:30013 -> 198.51.100.77:4444
2025-09-14 08:11:42 ALLOW TCP 198.51.100.77:56342 -> 10.0.0.50:25
2025-09-14 11:06:25 ALLOW TCP 198.51.100.77:58983 -> 10.0.0.51:25
2025-09-14 11:27:06 ALLOW TCP 198.51.100.77:51744 -> 10.0.0.60:80
2025-09-14 12:52:33 ALLOW TCP 198.51.100.77:54619 -> 10.0.0.50:8080
```

confirmed that data sent

```
2025-09-24 15:00:00 [**] [1:2002034:1] ET INFO Possible HTTP POST Large Upload [**]
al Data Exfiltration] [Priority: 2] {TCP} 10.0.0.51:40034 -> 198.51.100.77:80
2025-09-24 19:00:00 [**] [1:2002035:1] ET INFO Possible HTTP POST Large Upload [**]
al Data Exfiltration] [Priority: 2] {TCP} 10.0.0.51:40035 -> 198.51.100.77:80
2025-09-24 23:00:00 [**] [1:2002036:1] ET INFO Possible HTTP POST Large Upload [**]
al Data Exfiltration] [Priority: 2] {TCP} 10.0.0.51:40036 -> 198.51.100.77:8080
2025-09-25 03:00:00 [**] [1:2002037:1] ET INFO Possible HTTP POST Large Upload [**]
al Data Exfiltration] [Priority: 2] {TCP} 10.0.0.51:40037 -> 198.51.100.77:80
```

Task 8 - Conclusion

In this room, you learned that an enterprise network is not just a collection of computers and servers. It's an ecosystem with critical components like firewalls, Active Directory, application servers, file/database servers, endpoints, and wireless access points.

- The network perimeter acts as the boundary between trusted and untrusted networks.
- Attackers test this boundary constantly with port scans, brute-force attempts, and exploits against exposed services.
- Monitoring the perimeter allows SOC analysts to detect these early signs before attackers can move deeper into the network.

As a security analyst, our role is to:

- Recognize normal vs suspicious network traffic.
- Escalate unusual activity to higher SOC tiers.
- Understand how each network component fits into the bigger picture of enterprise defense.